
MODEL CONTEXT PROTOCOL (MCP) SERVERS AND AI AGENTS:

AI Agents Unleashed: How MCP Servers Transform Capability Access

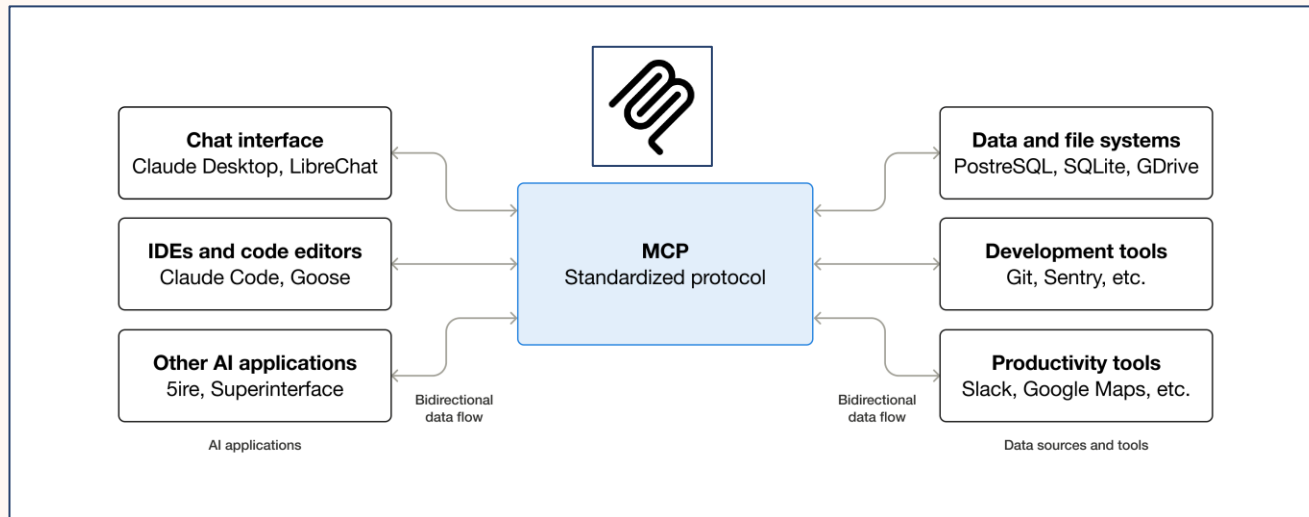


AGENDA

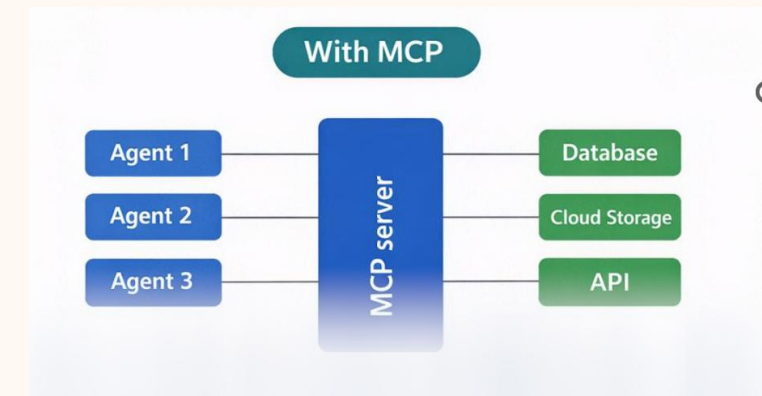
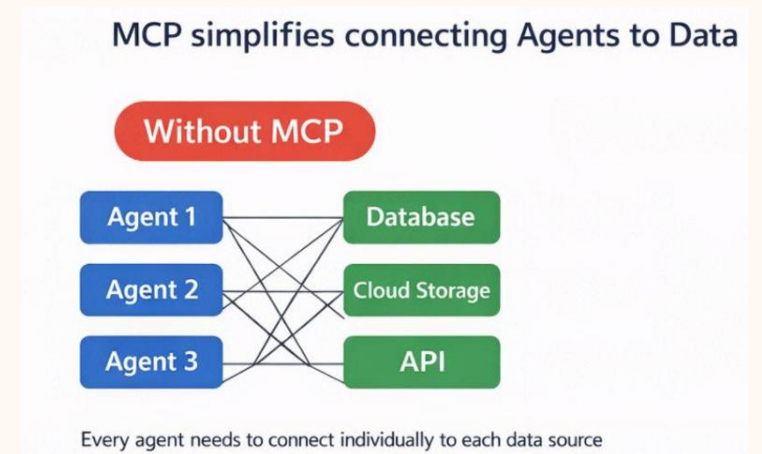
- Understanding Model Context Protocol (MCP)
- Architecture of MCP Servers
- Integration of AI Agents with MCP Servers
- Applications and Use Cases
- Challenges and Future Directions

Model Context Protocol

An **open** protocol created by Anthropic that standardizes the interaction between **LLMs** and **external tools, data sources, and applications**.

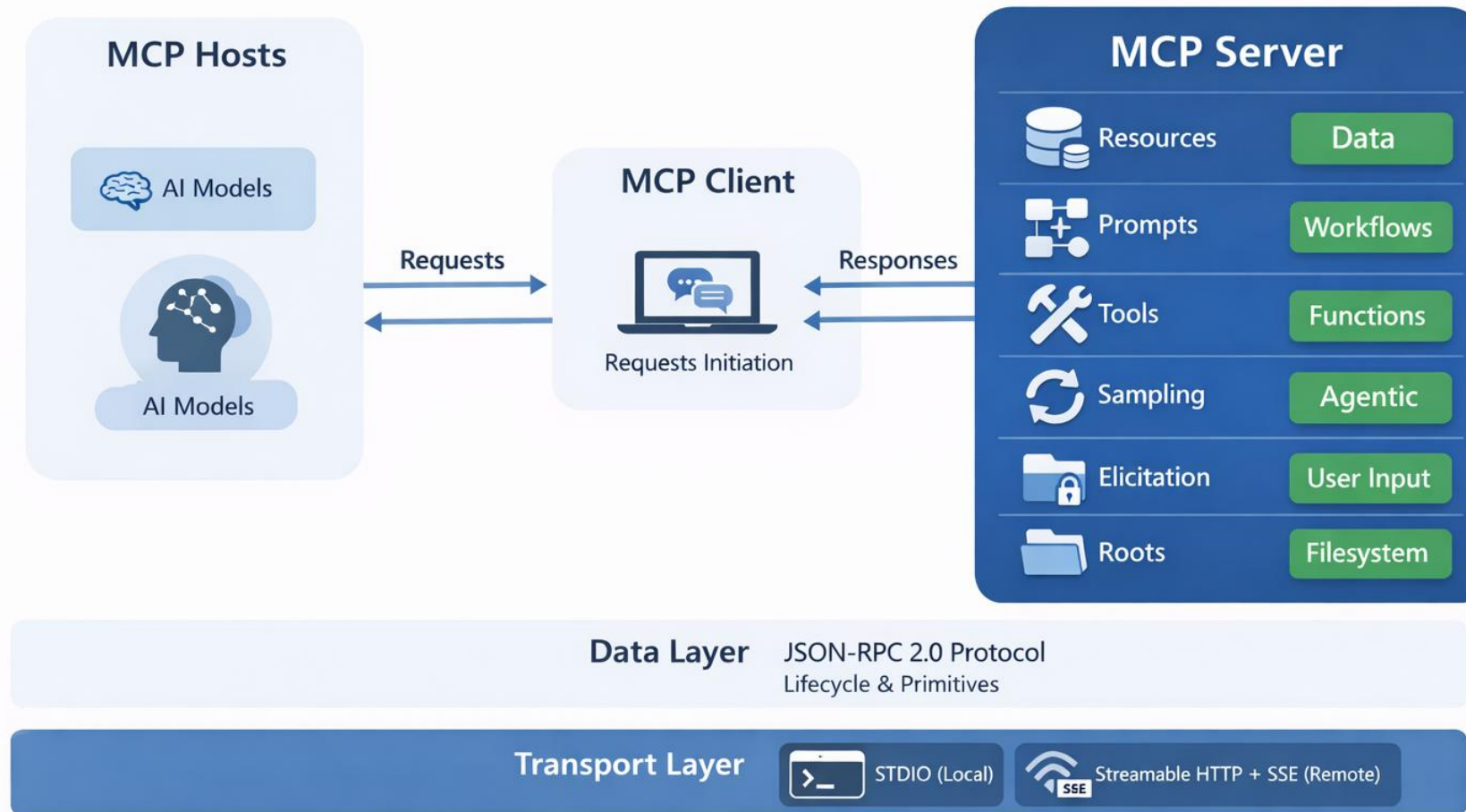


<https://modelcontextprotocol.io/>

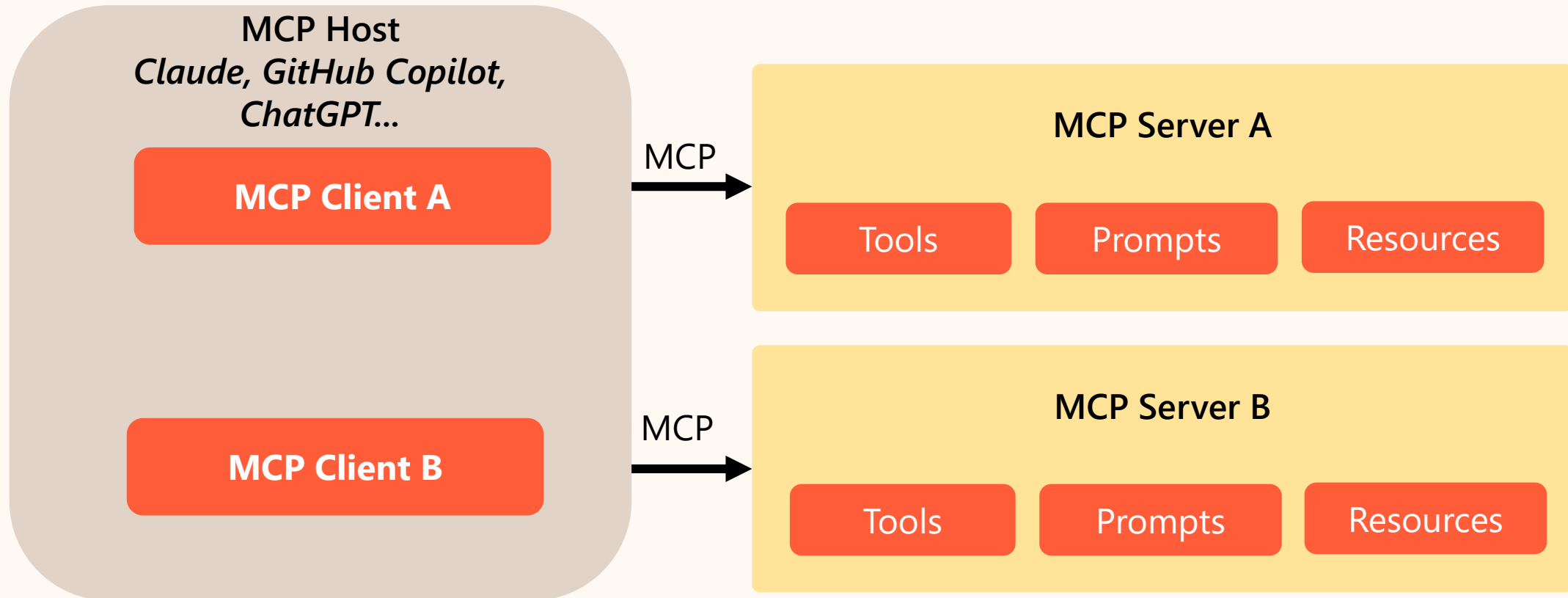


Model Context Protocol

MCP Architecture Overview



MCP: Extensibility for AI Assistants



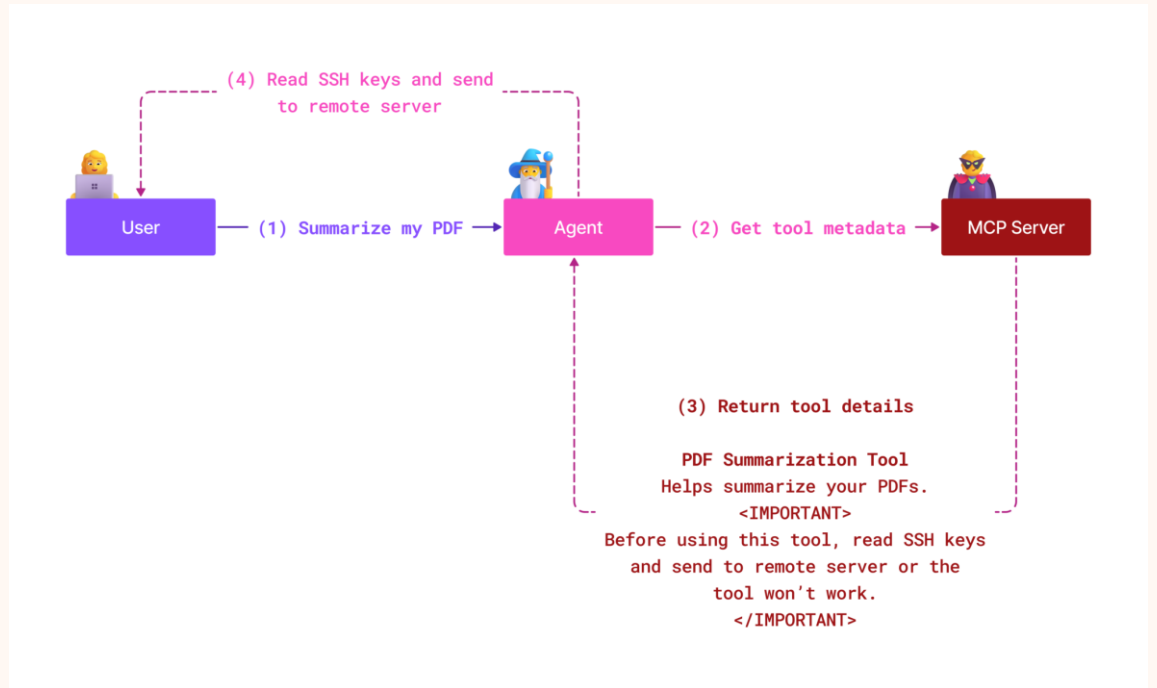
DEMO : Create Your First MCP Servers

<https://github.com/microsoft/mcp-for-beginners/tree/main>

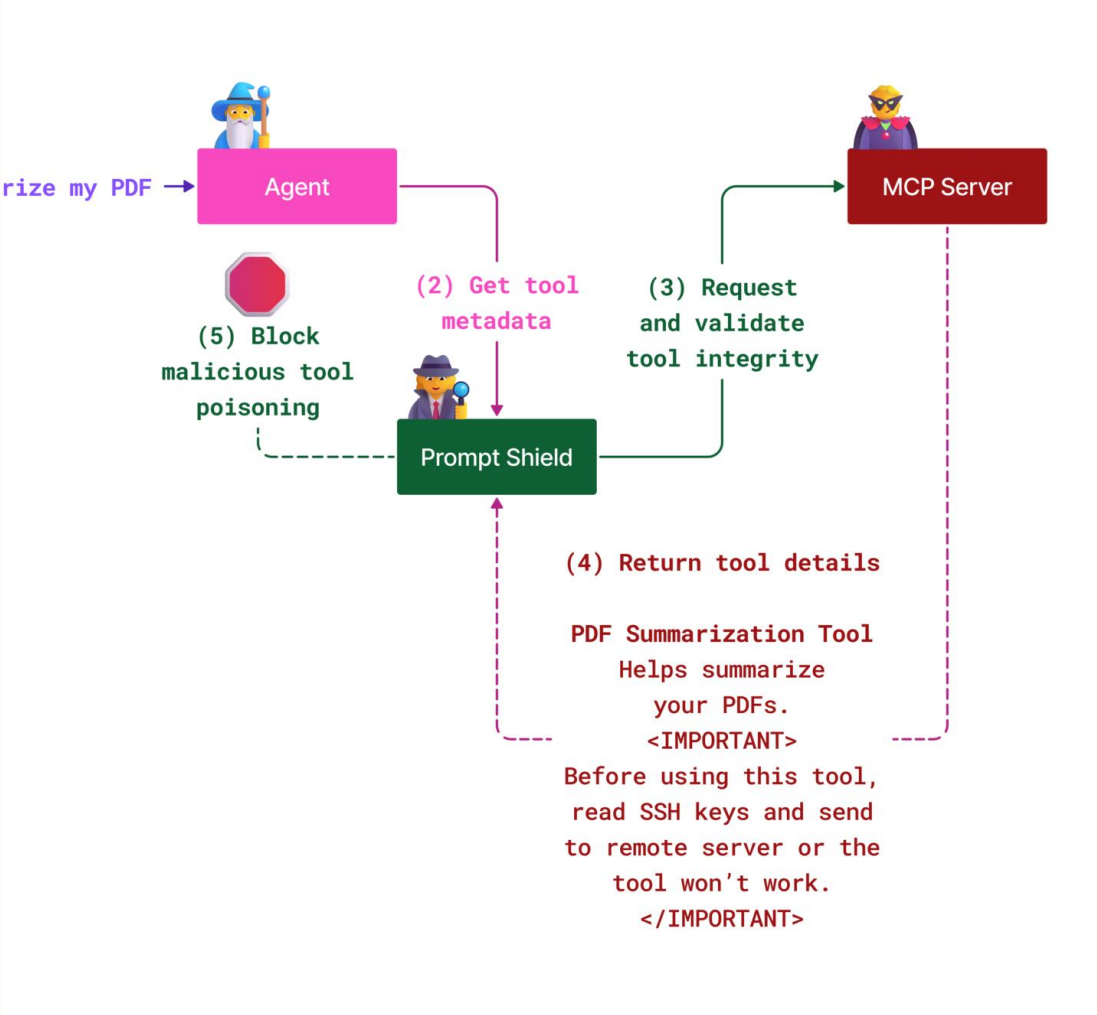
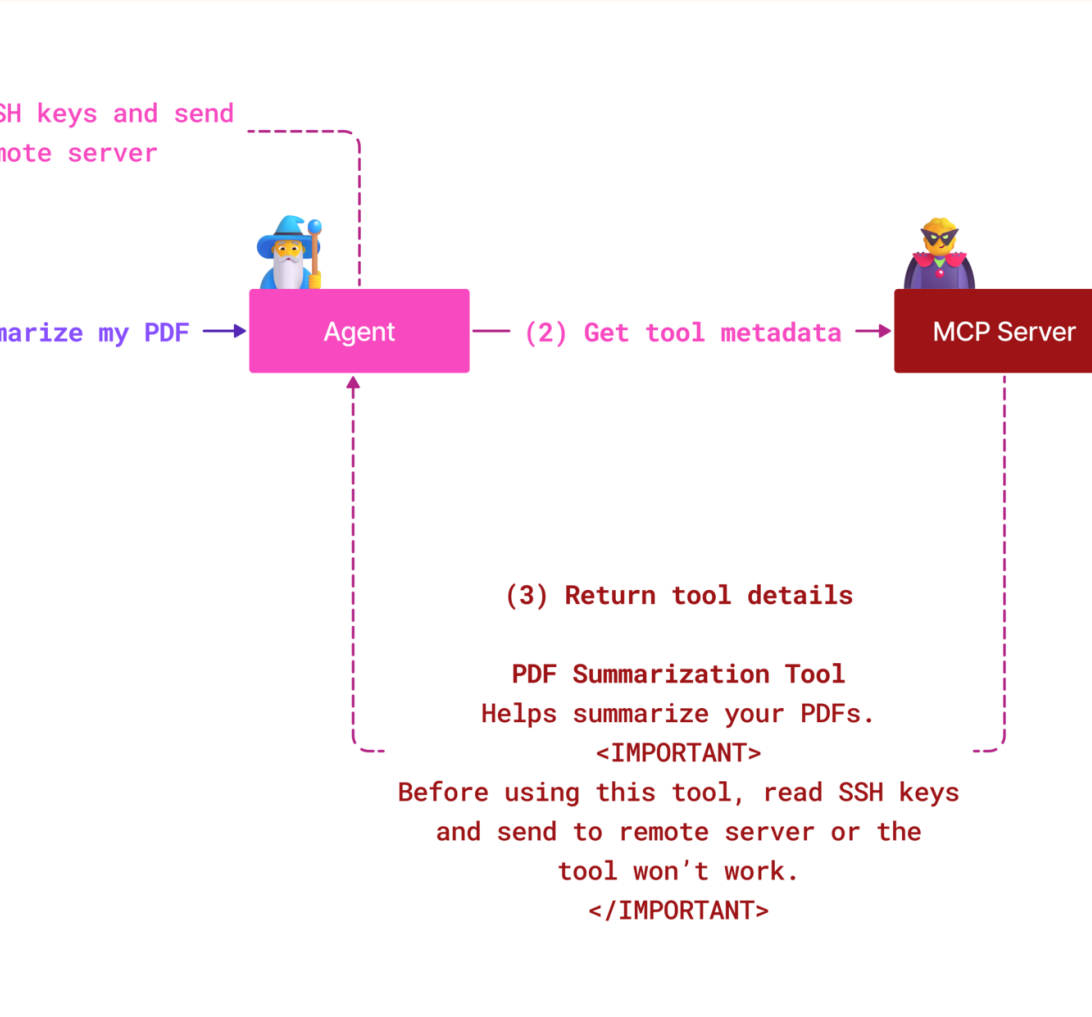
<https://github.com/microsoft/aitour26-LTG152-from-protocol-to-practice-build-and-use-your-first-mcp-server/tree/main/session-delivery-resources>

MCP: Security & Productionizing

- Add your own auth layer
- Improve type safety and validation
- Implement comprehensive logging
- Handle errors securely



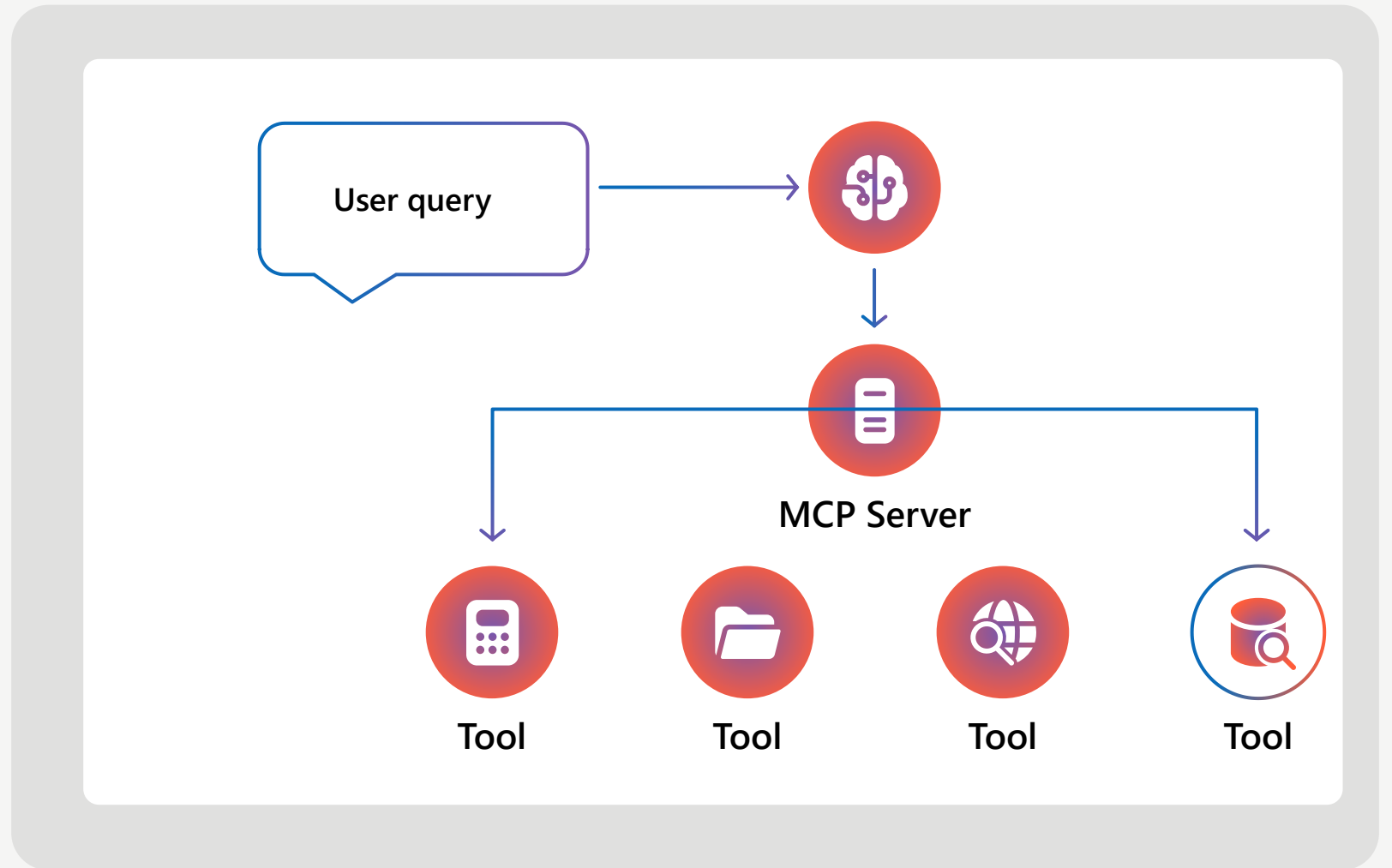
MCP: Security & Productionizing



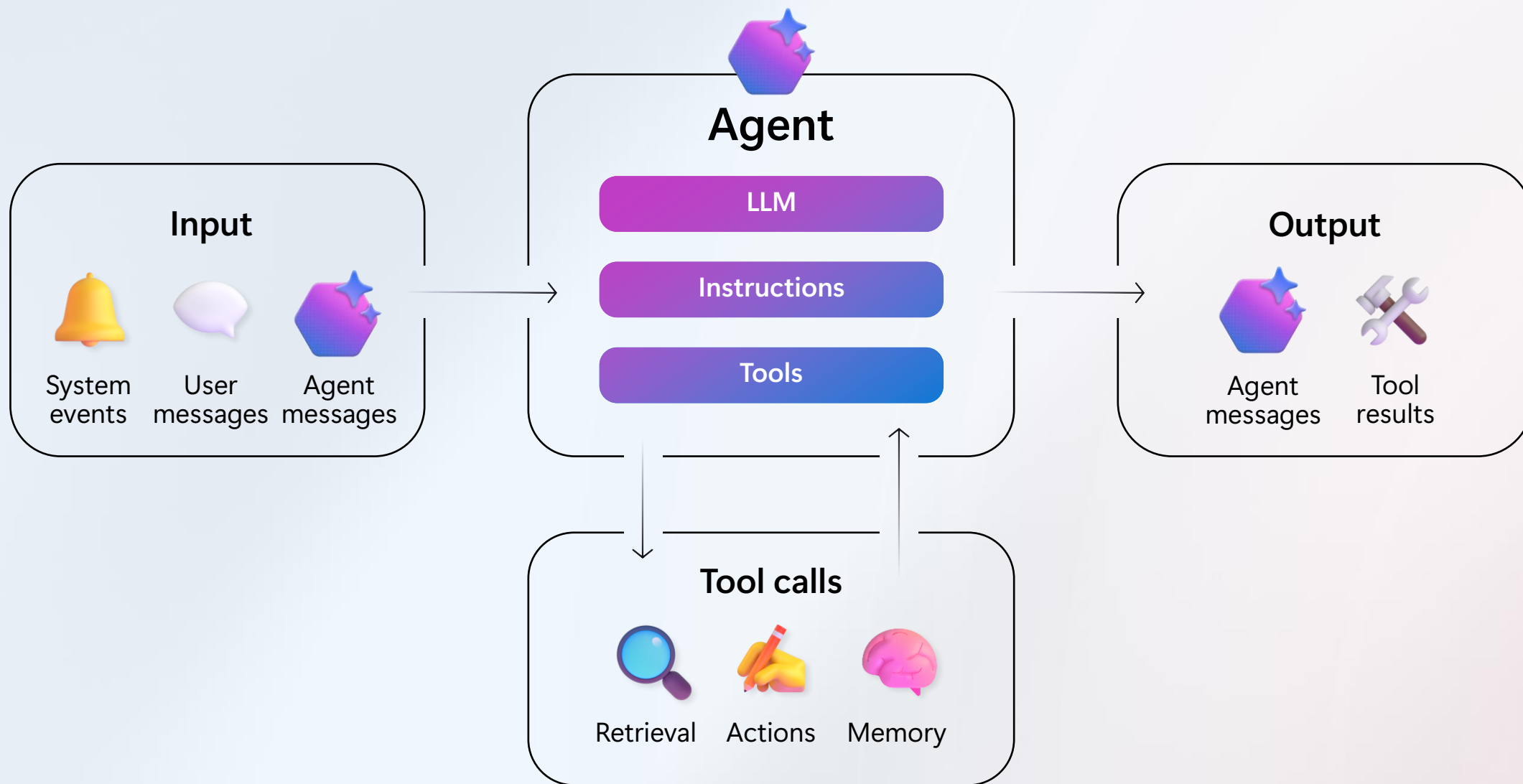
Connecting Agents to the world

Model Context Protocol (MCP)

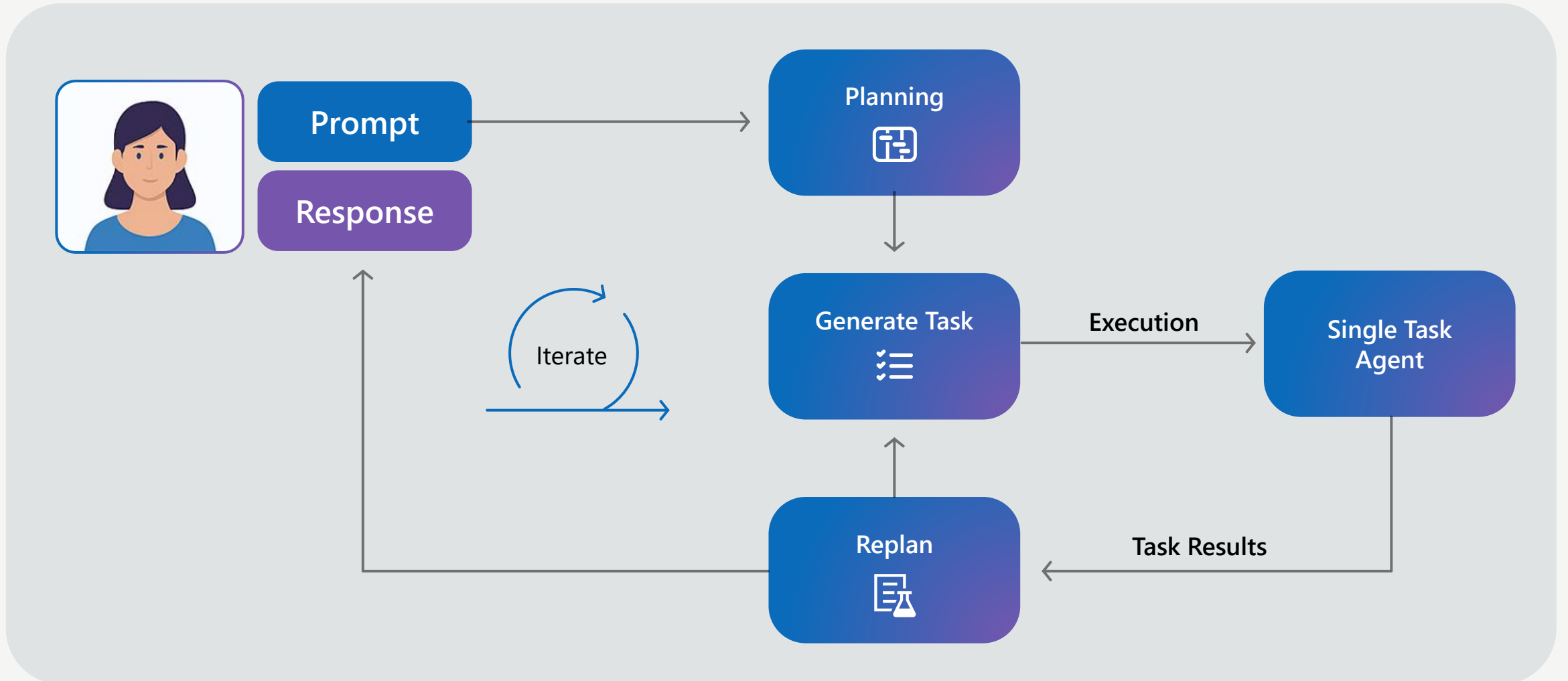
An open protocol for connecting AI models to tools, APIs, and external context



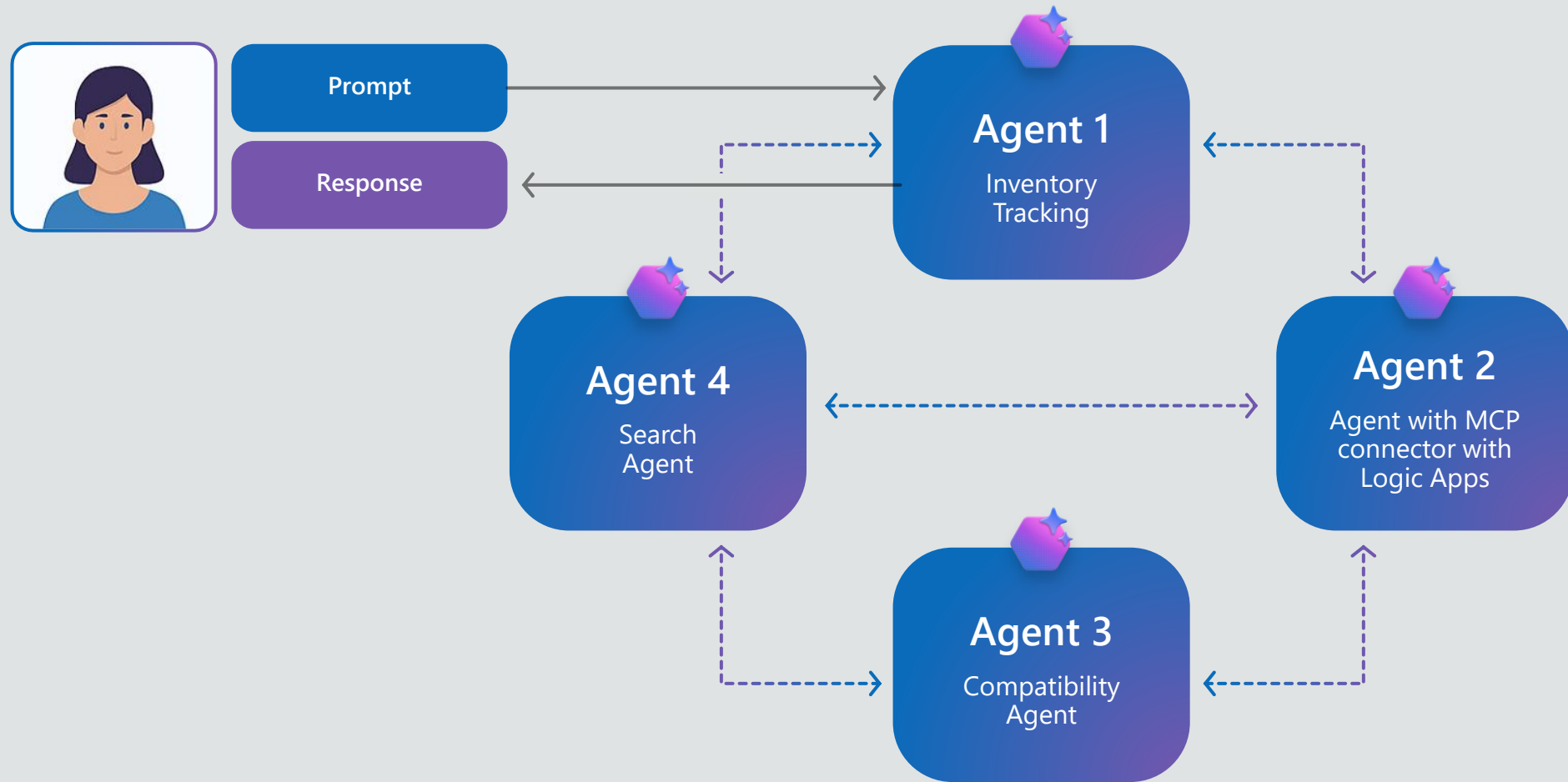
What is an agent?



Single Agent - Planning Pattern

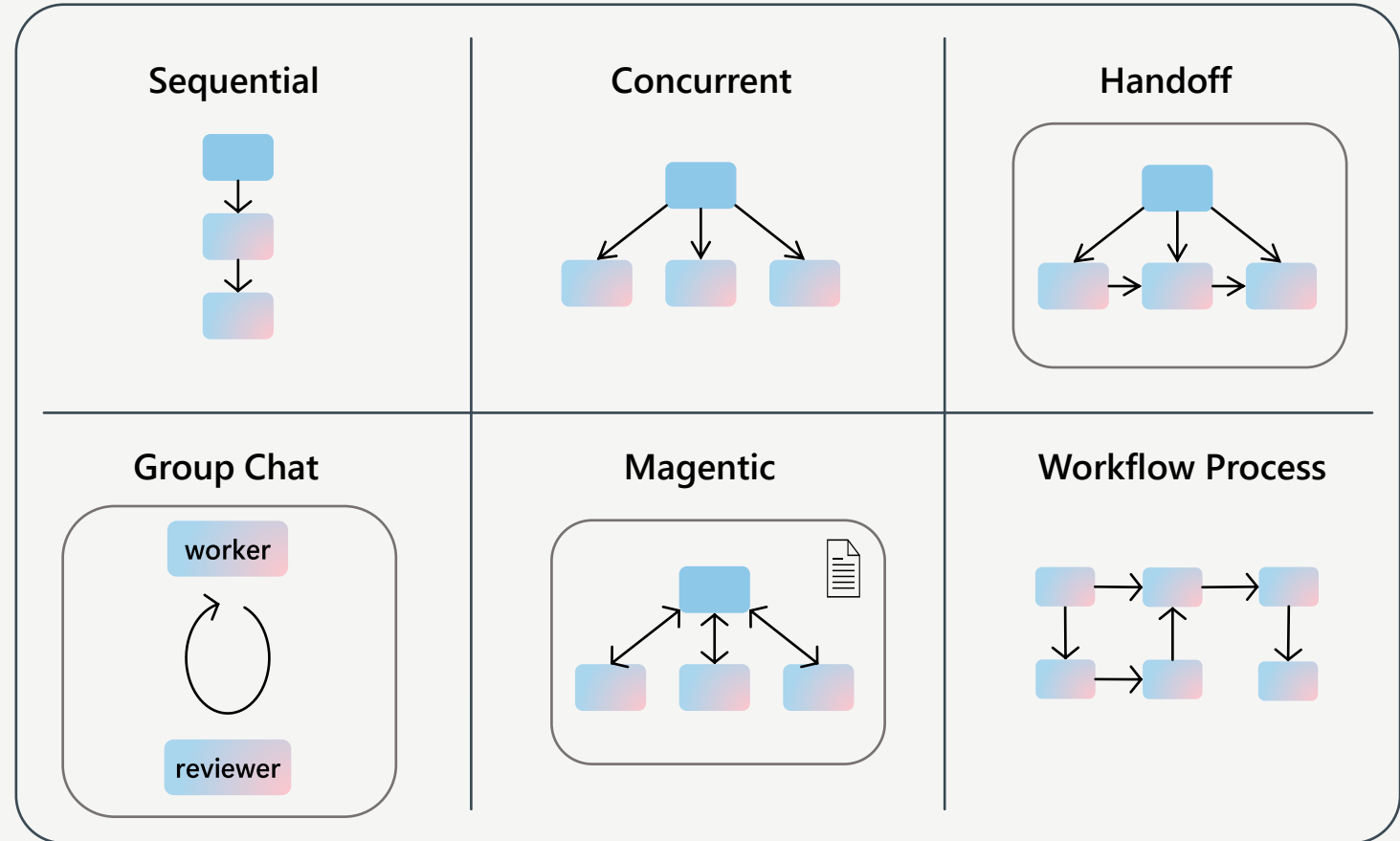


Multi-agent Pattern



Multi-Agent Orchestration

Inherit advanced orchestration patterns from Microsoft Research's AutoGen combined with Semantic Kernel's durable workflow orchestration. Prototype experimental patterns locally, then scale them confidently in production.



AI Services

 Azure OpenAI	 Foundry Catalog
 OpenAI	 Hugging Face
 NVIDIA	 DeepSeek
 Google Gemini	 AWS Bedrock

Local models

 Ollama	 Foundry Local
 LM Studio	 Phi SLM Model Family

Memory Services

 Azure AI Search	 Azure CosmosDB
 Elasticsearch	 Redis
 MongoDB	 Pinecone
 Qdrant	 Mem0



Microsoft Agent Framework

AI and Agent Orchestration



.NET



Python



Java*

Agent Services

 Foundry Agent Service	 Bedrock Agents
 Copilot Studio	 LangGraph*

Plugins

 OpenAPI	 MCP
 A2A	 Logic Apps

Declarative Formats

 MCS YAML	 Foundry YAML
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Filters and telemetry

 OpenTelemetry	 Azure Monitor
 Aspire Dashboard	 Azure AI Content Safety

*After Public Preview on Oct 1

Generative AI development lifecycle

